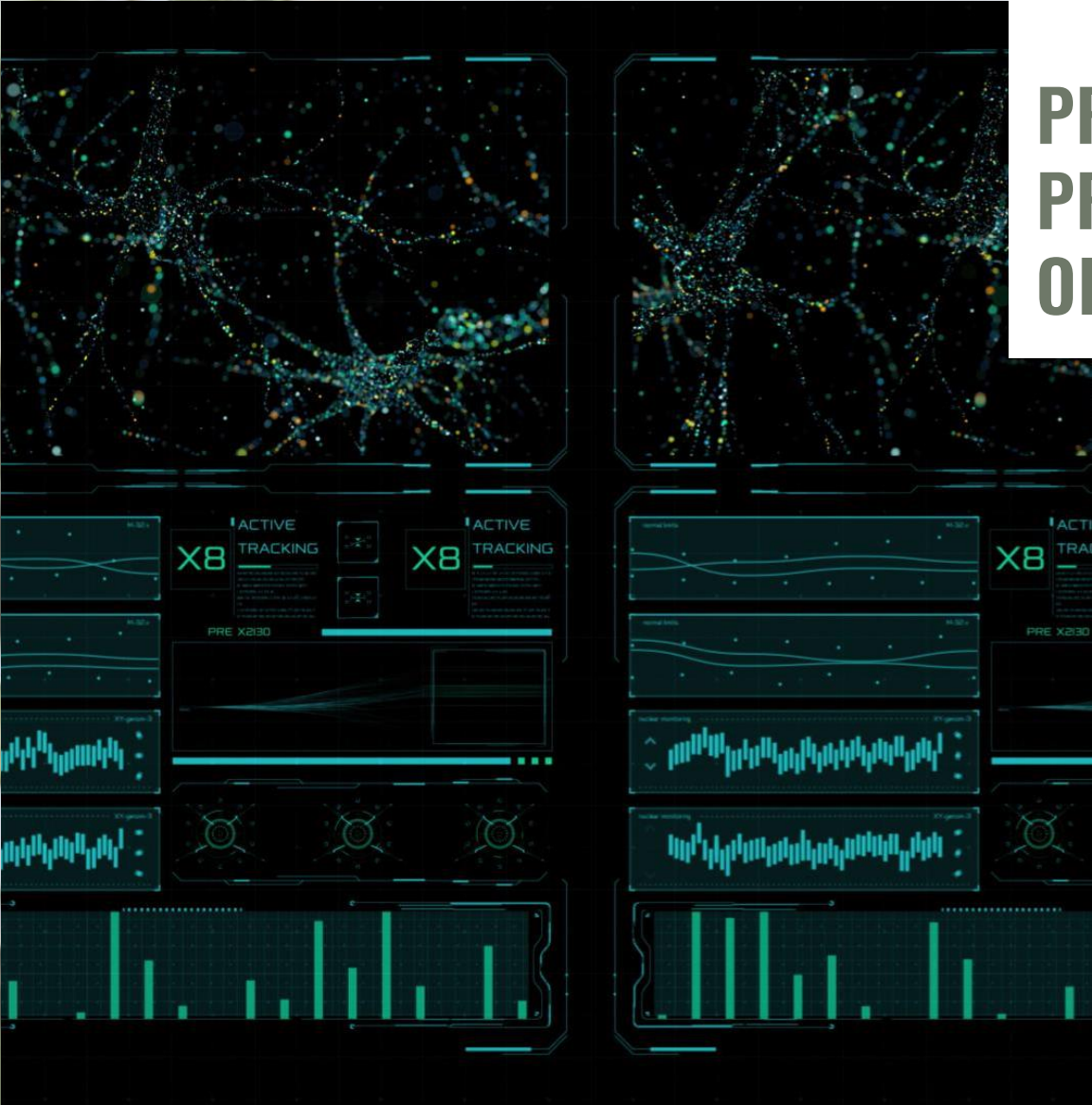


EXPLAINABLE HEALTHCARE ANALYTICS FRAMEWORK FOR LIVER CIRRHOSIS PREDICTION – DEMO PRESENTATION

INNOVATIVE TOOLS FOR INTERPRETING LIVER
DISEASE RISK ACCURATELY

PROJECT INTRODUCTION AND CONTEXT



PRESENTER, PROJECT TITLE, PROBLEM STATEMENT, AND OBJECTIVES

Project Overview

The project develops an explainable healthcare analytics framework for liver cirrhosis prediction using visual analytics and machine learning techniques.

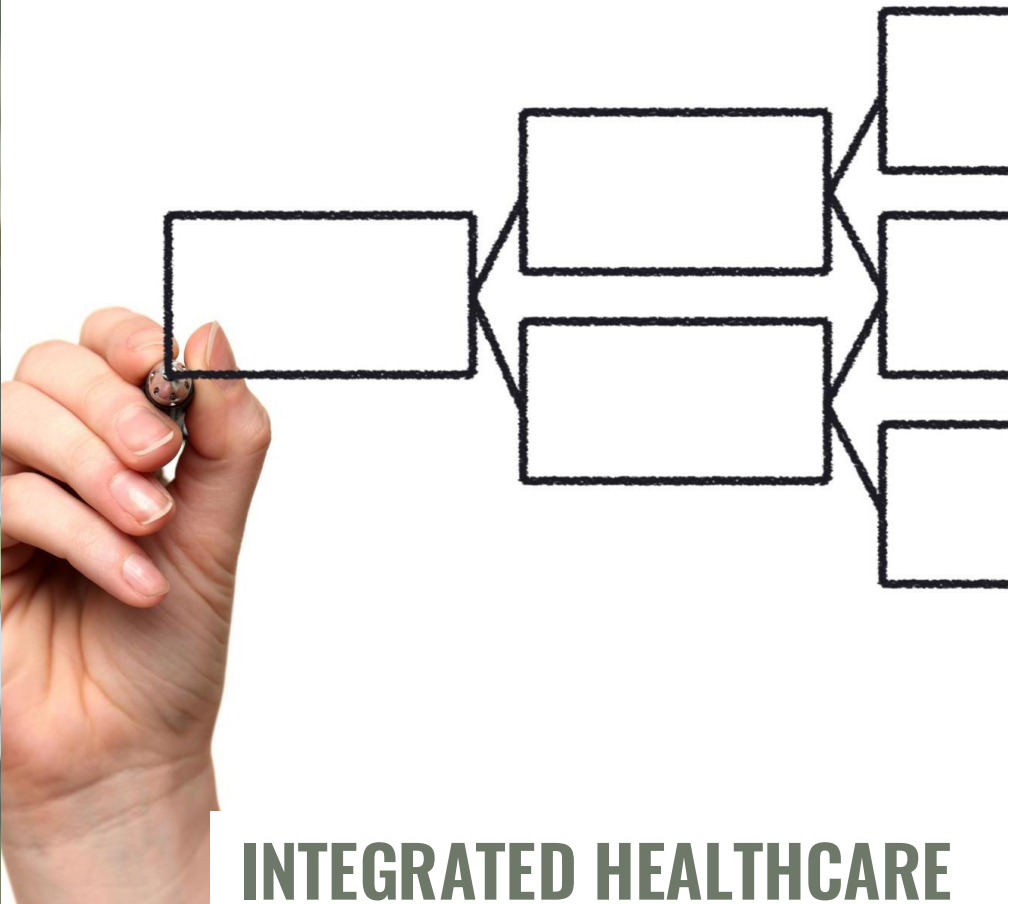
Problem Statement

Machine learning models often lack interpretability, reducing clinician trust and raising ethical concerns in healthcare environments.

Project Objectives

Focuses on transparency, accountability, and human-centred design by integrating exploratory data analysis, interpretable models, and visual dashboards.

SYSTEM OVERVIEW AND ARCHITECTURE



INTEGRATED HEALTHCARE ANALYTICS FRAMEWORK OVERVIEW

End-to-End Analytical Pipeline

The framework integrates data ingestion, exploration, modelling, and explainability into a seamless healthcare analytics pipeline.

Visual Analytics with Dashboards

Interactive Tableau dashboards enable exploration of demographic data, biomarkers, and disease severity patterns visually.

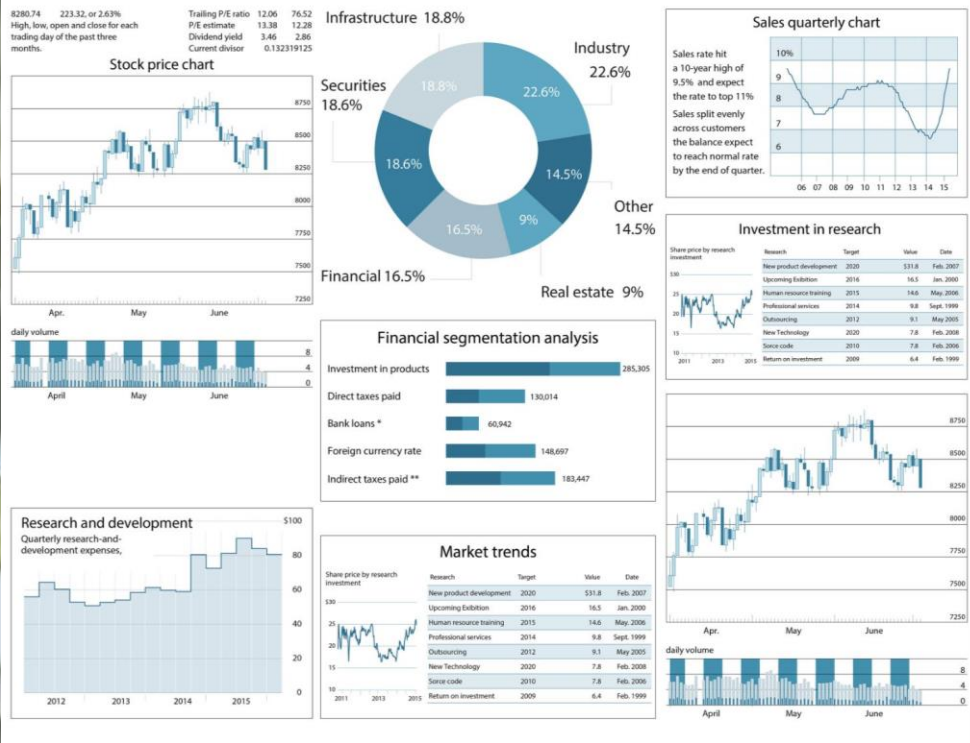
Predictive Modelling Using Decision Trees

RapidMiner decision tree models provide transparent and interpretable predictions suitable for healthcare contexts.

Explainable and Responsible AI

The framework supports explainable reasoning and aligns with ethical AI principles to aid transparent healthcare decisions.

SYSTEM AND PRODUCT DEMONSTRATION



LIVE DEMONSTRATION OF THE HEALTHCARE ANALYTICS SYSTEM

Interactive Data Exploration

Users explore healthcare data via dashboards, filtering demographics and clinical biomarkers interactively.

Visual Analytics for Insights

Visual analytics reveal complex healthcare relationships that are hard to interpret from raw data.

Predictive Modelling Workflow

Demonstration of transparent data preprocessing, training, and evaluation in a visual workflow environment.

Integrated System Understanding

The demo guides users narrative-driven through system features, highlighting technical and healthcare relevance.

PREDICTIVE MODELLING APPROACH



DECISION TREE MODELLING FOR EXPLAINABLE HEALTHCARE PREDICTION

Explainable Decision Trees

Decision trees provide transparent, rule-based pathways that clinicians can easily understand and justify.

Healthcare Variables Evaluated

Key clinical variables like age, bilirubin, albumin, and platelet counts guide prediction decisions in the model.

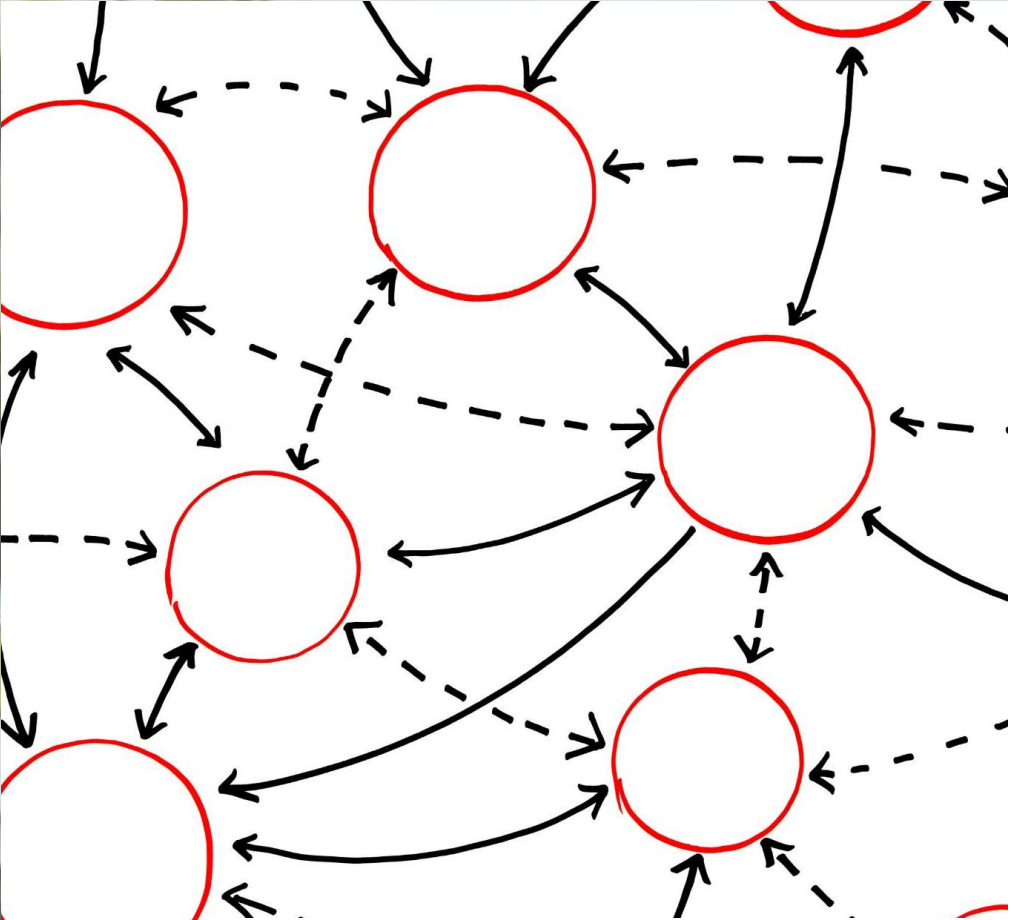
RapidMiner Workflow

The modelling process uses RapidMiner to visually represent preprocessing, model training, and evaluation steps.

Accuracy vs Interpretability

The model balances predictive accuracy with interpretability to meet healthcare usability and ethical standards.

CODE AND WORKFLOW DEMONSTRATION



EXPLANATION OF KEY CODE AND RAPIDMINER WORKFLOWS

Workflow-Based Implementation

Focuses on structured workflows in RapidMiner instead of large raw code volumes for clarity and reproducibility.

Data Handling and Partitioning

Covers dataset import, handling categorical and numerical variables, target variable assignment, and data partitioning.

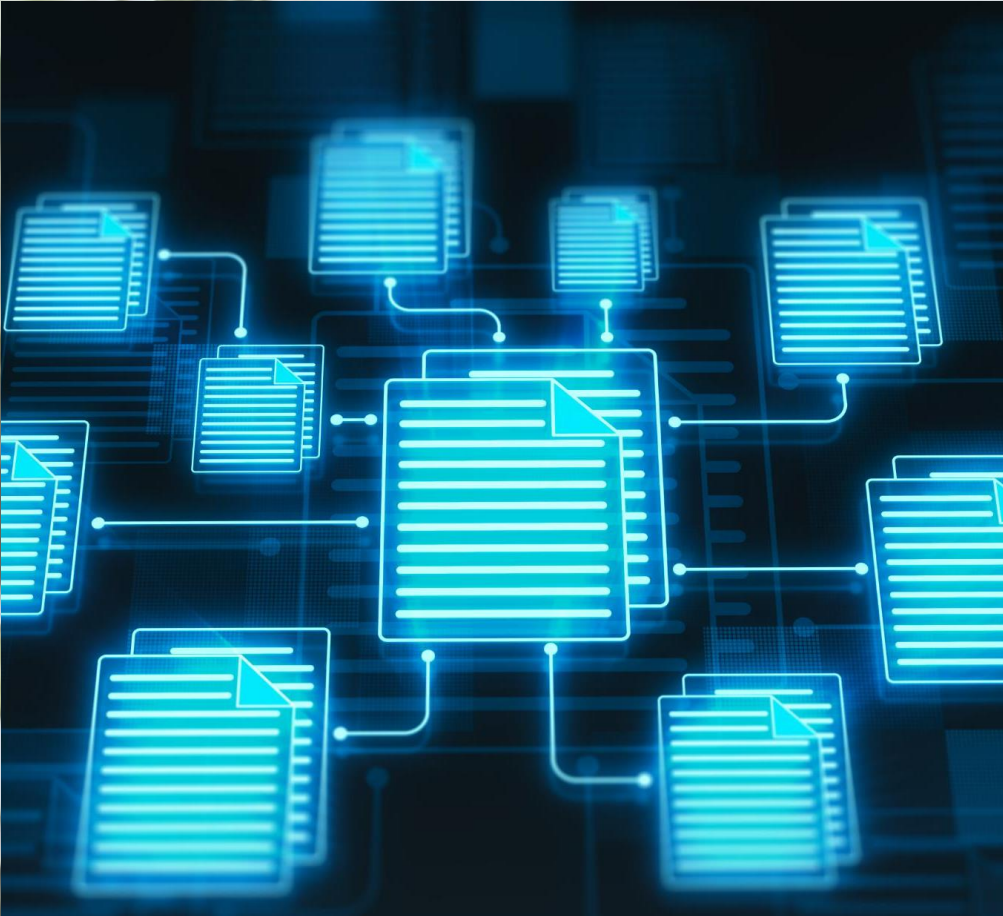
Decision Tree Visualization

Highlights use of decision tree operators to generate interpretable and visualized models for better understanding.

Transparency and Explainability

Emphasizes how transparent workflows enhance understanding and reproducibility over complex scripted code.

REPOSITORY AND REPRODUCIBILITY



PROJECT REPOSITORY AND ACCESS TO CODE AND RESOURCES

Repository Contents Overview

The repository hosts code, workflows, dashboards, and documentation essential for project transparency and usability.

Support for Reproducibility

Access to the repository enables others to review, run, and extend the project, ensuring reproducibility.

Best Software Practices

Clear folder structures, descriptive naming, and README files promote professional software engineering standards.

Transparency and Credibility

Open access to the repository reassures assessors of the project's credibility and ease of independent review.

RESULTS AND DISCUSSION



PREDICTIVE RESULTS, INTERPRETATION, AND SYSTEM PERFORMANCE

Key Predictive Variables

Biomarkers like bilirubin and albumin, along with age, significantly influence cirrhosis severity classification outcomes.

Model Interpretability

The decision tree model produces clear, interpretable structures aiding understanding of classification decisions.

Visual Analytics Role

Visual analytics enhances result interpretation by showing variable distributions and relationships contextually.

Limitations and Evaluation

Limitations include dataset size, representativeness, and lack of clinical deployment, encouraging critical performance evaluation.

LIMITATIONS AND FUTURE WORK



SYSTEM LIMITATIONS AND OPPORTUNITIES FOR EXTENSION

Current System Limitations

System relies on small, disease-specific data and lacks longitudinal studies and clinical validation.

Analytical Strengths and Purpose

System demonstrates integration and explainability but is not designed for clinical deployment.

Future Work Opportunities

Use larger diverse datasets, incorporate healthy data, and explore explainable AI tools like SHAP and LIME.

Extensions and Ethical Considerations

Potential for real-time integration, cloud analytics, preventative risk tools, with focus on ethics and governance.

CONCLUSION AND CLOSING



KEY TAKEAWAYS AND FINAL REMARKS

Integrated Healthcare Analytics

The project developed a unified framework combining visual analytics and interpretable machine learning approaches for healthcare.

Importance of Explainability

Explainability and transparency are critical for building trust in AI systems used in healthcare environments.

Academic Integration

The framework integrates multiple advanced computer science domains into a cohesive solution showcasing academic rigor.

Future Potential and Closure

The presentation concludes with project relevance, limitations, and future opportunities for healthcare AI advancements.